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Smart Ambulance Rapid Response System

Problem : Ambulances are delayed by traffic congestion and poor coordination between dispatch, crew, and hospitals — costing critical minutes in an emergency.

PROBLEM
STATEMENT ID

HE-02

THEME

Health care

PS CATEGORY

Software

TEAM

Bio Pulse

Faster Ambulance = More Lives Saved

HOW IT WORKS

Proposed Solution

SDG GOALS ADDRESSED

3.

Good Health and
Well-being

9.

Industry, Innovation
and Infrastructure

11.

Sustainable Cities
and Communities

Automatic Route Clearance

The dispatch app automatically requests priority traffic-signal changes along the ambulance's route, clearing a faster path through congestion.

Emergency Alert App

Patient details are captured through the app or a phone call and sent ahead to the receiving hospital, so staff can prepare before the ambulance arrives.

Response Workflow

START

1 Call & Open App

2 Dispatch Request Sent

3 Assign Nearest Ambulance

4 Crew Gets Alert on Tablet

5 Ambulance Drives With GPS

CONTINUED

1 Track Live Location

2 Ambulance Reaches Patient

3 Paramedics Provide Care

4 Hospital Updated & Prepared

5 Transport Patient to Hospital

END

Technical Approach

1 Frontend & Backend

HTML, CSS, JS, UI / UX, Node.js

2 AI Source

Private AI Agent — secure, cloud-synced access

3 AI / ML

ETA estimation and dispatch-assignment logic

x

Feasibility and Viability

Main Problem	Solution
Ambulance Response Delay	Smart Dispatch
Traffic & Route Delay	AI-Based Route Optimization
Patient–Ambulance–Hospital Coordination	Hospital Pre-Alert & Tracking

OUTCOMES

Impact and Benefits

IMPACT

- Reduces emergency response time
- Finds the best-matched ambulance
- Improves traffic management along the route
- Gives an accurate estimated time of arrival
- Enables real-time location tracking

BENEFITS

- Faster emergency response
- Optimized ambulance assignment
- Best route navigation
- Real-time live updates
- Earlier hospital preparation

OUR GOAL

To create a fast emergency-response system that ensures the right help reaches the right person at the right time.

Thank You

BioPulse — Smart Ambulance Rapid Response System